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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/595,715	11/17/2009	Jens Steen Olsen	Q115476	5595
23373 7590 04/13/2012 SUGHRUE MION, PLLC 2100 PENNSYLVANIA AVENUE, N.W. SUITE 800 WASHINGTON, DC 20037			EXAMINER HOFFMAN, SUSAN COE	
			ART UNIT 1655	PAPER NUMBER
			NOTIFICATION DATE 04/13/2012	DELIVERY MODE ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

USPTO@sughrue.com
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Office Action Summary	Application No.	Applicant(s)	
	12/595,715	OLSEN, JENS STEEN	
	Examiner	Art Unit	
	Susan Hoffman	1655	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --
Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 30 January 2012.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 5) ☒ Claim(s) 36-55 is/are pending in the application.
- 5a) Of the above claim(s) 47-55 is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☒ Claim(s) 36-46 is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

Application Papers

- 10) ☒ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 13 October 2009 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
 Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
 Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 12) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 13) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. ____.
3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).
- * See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- 1) ☒ Notice of References Cited (PTO-892)
- 2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948)
- 3) ☒ Information Disclosure Statement(s) (PTO/SB/08)
 Paper No(s)/Mail Date 1/13/10
- 4) ☐ Interview Summary (PTO-413)
 Paper No(s)/Mail Date. ____
- 5) ☐ Notice of Informal Patent Application
- 6) ☐ Other: ____

DETAILED ACTION

1. Claims 36-55 are currently pending.
2. A third-party submission has been filed under 37 CFR 1.99 on June 29, 2010 in the published application.

To ensure that a third-party submission does not amount to a protest or pre-grant opposition, 37 CFR 1.99 does not permit the third party to have the right to insist that the examiner consider any of the patents or publications submitted. Furthermore, if the submission or part of the submission is not in compliance with 37 CFR 1.99, that noncompliant submission or part thereof will not be entered in the application file. Therefore, unless the examiner clearly cites a patent or publication on form PTO-892, Notice of References Cited and such reference is used in a rejection or its relevance is actually discussed during prosecution, consideration by the examiner of any patent or publication submitted in a third-party submission cannot be presumed.

If the applicant wants to ensure that the information in a third-party submission is considered by the examiner, the applicant should submit the information in an IDS in compliance with 37 CFR 1.97 and 37 CFR 1.98. An individual who has a duty to disclose under 37 CFR 1.56 should also submit any material information contained in a third-party submission to the Office in an IDS in compliance with 37 CFR 1.97 and 37 CFR 1.98 to ensure such material information is properly disclosed to the examiner.

Election/Restrictions

3. Applicant's election without traverse of Group I, claims 36-46, in the reply filed on January 30, 2012 is acknowledged.
4. Claims 47-55 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected invention, there being no allowable generic or linking claim.
5. Claims 36-46 are examined on the merits.

Specification

6. The disclosure is objected to because of the following informalities: it does not contain a "Brief Description of the Drawings" section.

Appropriate correction is required.

Claim Rejections - 35 USC § 112

The following is a quotation of the first paragraph of 35 U.S.C. 112:

The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same and shall set forth the best mode contemplated by the inventor of carrying out his invention.

7. Claims 36-46 are rejected under 35 U.S.C. 112, first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor(s), at the time the application was filed, had possession of the claimed invention. This is a "written description" rejection, rather than an enablement rejection under 35 U.S.C. 112, first paragraph. Applicant is directed to the Guidelines for the Examination of

Patent Applications Under the 35 U.S.C. 112, § 1 "Written Description" Requirement, Federal Register, Vol. 66, No. 4, pages 1099-1111, Friday January 5, 2001.

The claims are drawn to an extract "**obtainable**" and a fraction "**obtainable**" according to the procedures set forth in claims 36-43. The use of "obtainable" does not limit the claims to extracts and fractions made according to the procedures set forth in the claims. Thus, the claims can be interpreted to read on any extract and any fraction from any extract due to the broad scope of "obtainable". However, the specification only discloses making extracts and fractions from *Trigonella foenum-graecum* using the procedures set forth in claims 36-43 and in the specification at pages 2-5 and Examples 1 and 13-15. In analyzing whether the written description requirement is met for genus claims, it is first determine whether a representative number of species have been sufficiently described. In this case, only extracts and fractions from *T. foenum-graecum* using the procedures set forth in claims 36-43 and in the specification at pages 2-5 and Examples 1 and 13-15 are described. Applicant has only demonstrated possession of these particular extracts and extract fractions. This limited information is not deemed sufficient to reasonably convey to one skill in the art that applicant was in possession of all potential extracts and extract fractions at the time the application was filed. As discussed in MPEP section 2163:

The written description requirement for a claimed genus may be satisfied through sufficient description of a representative number of species... A "representative number of species" means that the species which are adequately described are representative of the entire genus. Thus, when there is substantial variation within the genus, one must describe a sufficient variety of species to reflect the variation within the genus. The disclosure of only one species encompassed within a genus adequately describes a claim directed to that genus only if the disclosure "indicates that the patentee has invented species sufficient to constitute the gen[us]." See *Enzo Biochem*, 323 F.3d at 966, 63 USPQ2d at 1615; *Noelle v. Lederman*, 355 F.3d 1343, 1350, 69 USPQ2d 1508, 1514 (Fed. Cir. 2004) (Fed. Cir. 2004)...

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Since there are millions of potential extracts and extract fraction, there would be considerable variation within this plant family. The description of extracts and extract fractions from *T. foenum-graecum* within this large genus is not considered to adequately describe the entire genus. Thus, it is concluded that the written description requirement is not satisfied for the claimed genus.

The following is a quotation of the second paragraph of 35 U.S.C. 112:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

8. Claim 43 is rejected under 35 U.S.C. 112, second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which applicant regards as the invention. There is a lack of antecedent basis for "the solvent" in line 1 of the claim.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(b) the invention was patented or described in a printed publication in this or a foreign country or in public use or on sale in this country, more than one year prior to the date of application for patent in the United States.

9. Claims 36-39 and 43-46 are rejected under 35 U.S.C. 102(b) as being anticipated by Hardman (US 3,620,919).

This reference teaches a *T. foenum-graecum* extract obtained by incubating a mixture of the plant seeds in water over 3 hours; heating the mixture to reflux for up to 5 hours; and filtering

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the mixture to obtain a liquid extract. The solvent is removed by evaporation (see Example 3 and column 4, lines 44-46). The reference teaches that the extracts can be used as pharmaceutical compositions.

10. Claims 36, 37, 40 and 43-46 are rejected under 35 U.S.C. 102(b) as being anticipated by Lee (US 2005/0238738).

This reference teaches a *T. foenum-graecum* extract made by extracting the seeds in a first solvent at a temperature between 20°C and 90°C for up to about three hours. According to applicant's specification at page 3, incubation is performed at temperatures that overlap with this range. Thus, this step in the reference is considered to meet applicant's limitation that the mixture is "incubated." The reference teaches that the extract is then boiled to recover an extract. The extract is mixed with water to form a liquid. The final extract is concentrated by freeze-drying (see paragraphs 93 and 109). Freeze-drying includes a freezing step; thus, this step is considered to meet applicant's claim 41. The reference teaches that the composition is a pharmaceutical composition (see abstract).

11. Claims 36, 37, 39, 44-46 are rejected under 35 U.S.C. 102(b) as being anticipated by Ziya Al-Din Abdullah Al-Jaam'e-li-Mufradaat-al-Advia-wal-Aghzia, Vol. II (13th century AD), Matba Amra, Cairo, Egypt, 1874 AD, page 25.

This reference a pharmaceutical composition comprising an extract from *T. foenum-graecum* seed. The extract is made by soaking the seed in water overnight. Applicant's specification at page 3 states that the incubation step can take place at temperatures that encompass average room temperatures. Thus, this step in the reference is considered to meet

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applicant's incubation step. The soaked mixture is then heated to a boil to recover a liquid extract.

12. Claims 36, 37, 39, 44-46 are rejected under 35 U.S.C. 102(b) as being anticipated by Mohammad Akmal Khan Qaraabaadeen Azam wa Akmal (20th century AD), Matba Siddiqi, Delhi / Matba Mustafai, Delhi, 1909 AD, page 167.

This reference a pharmaceutical composition comprising an extract from T. foenum-graecum seed. The extract is made by soaking the seed in water overnight. Applicant's specification at page 3 states that the incubation step can take place at temperatures that encompass average room temperatures. Thus, this step in the reference is considered to meet applicant's incubation step. The soaked mixture is then heated to a boil to recover a liquid extract.

Claim Rejections - 35 USC § 102/103

The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

13. Claims 36-41 and 43-46 rejected under 35 U.S.C. 102(b) as anticipated by or, in the alternative, under 35 U.S.C. 103(a) as obvious over Hardman (US 3,620,919).

Claims 36-41 and 43-46 are product-by-process claims. Regarding product-by-process claims, note that MPEP § 2113 states that:

[w]hen the prior art discloses a product which reasonably appears to be either identical with or only slightly different than a product claimed in a product-by-process claim, a rejection based alternatively on either section 35 U.S.C. 102 or 35 U.S.C. 103 of the statute is appropriate. ...A lesser burden of proof is required to make out a case of *prima facie* obviousness for product-by-process claims because of their peculiar nature than when a product is claimed in the conventional fashion. In *re Brown*, 59 CCPA 1063, 173 USPQ 685 (1972) ; In *re Fessmann*, 180 USPQ 324 (CCPA1974)... Once the examiner provides a rationale tending to show that the claimed product appears to be the same or similar to that of the prior art, although produced by a different process, the burden shifts to applicant to come forward with evidence establishing an unobvious difference between the claimed product and the prior art product. In *re Marosi*, 710 F.2d 798, 802, 218 USPQ 289, 292 (Fed. Cir. 1983).

The teachings of this reference are discussed above. The reference discloses extract which appears to be identical to the presently claimed extract, based on the fact that the both the reference extract and the claimed extract are from the same plant, are extracted in a similar manner, and have pharmaceutical properties. Consequently, the claimed extract appears to be anticipated by the reference.

However, even if the reference extract and the claimed extract are not one and the same and there is, in fact, no anticipation, the reference extract would, nevertheless, have rendered the claimed extract obvious to one of ordinary skill in the art at the time the claimed invention was made in view of the clearly close relationship between the extracts as evidenced by their shared characteristics.

Thus the claimed invention as a whole was clearly *prima facie* obvious especially in the absence of sufficient, clear, and convincing evidence to the contrary.

14. Claims 36-46 rejected under 35 U.S.C. 102(b) as anticipated by or, in the alternative, under 35 U.S.C. 103(a) as obvious over Lee (US 2005/0238738).

The teachings of this reference are discussed above. In addition to the extraction step discussed above, the reference also teaches performing HPLC on the extracts using gradient solvents such as acetonitrile (AcN) and water (see paragraph 120 and 121). Thus, the reference discloses extracts and fractions which appear to be identical to the presently claimed extract and fraction, based on the fact that the both the reference extract and the claimed extract are from the same plant, are extracted in a similar manner, and have pharmaceutical properties. Consequently, the claimed extract appears to be anticipated by the reference.

However, even if the reference extract and the claimed extract are not one and the same and there is, in fact, no anticipation, the reference extract would, nevertheless, have rendered the claimed extract obvious to one of ordinary skill in the art at the time the claimed invention was made in view of the clearly close relationship between the extracts as evidenced by their shared characteristics.

Thus the claimed invention as a whole was clearly *prima facie* obvious especially in the absence of sufficient, clear, and convincing evidence to the contrary.

Claim Rejections - 35 USC § 103

15. Claims 36, 40 and 41 are rejected under 35 U.S.C. 103(a) as being unpatentable over Hardman (US 3,620,919) in view of Lee (US 2005/0238738).

As discussed above, Hardman teaches a *T. foenum-graecum* extract obtained by incubating a mixture of the plant seeds in water over 3 hours; heating the mixture to reflux for up to 5 hours; and filtering the mixture to obtain a liquid extract. The solvent is removed by evaporation. The reference also teaches an additional extraction step using Soxhlet (see Example

3 and column 4, lines 44-46). Soxhlet uses heat during the extraction. The reference does not teach freezing the material before or after the heat treatments. The reference does teach removing solvent after each heating step. Lee demonstrates that it was known in the art at the time of the invention to use freeze-drying to remove solvents from extraction solutions (see paragraph 109). Freeze-drying includes a freezing step. Thus, an artisan of ordinary skill would be reasonably expect that freeze-drying steps could be used during the extraction procedure taught by Hardman to remove the solvent as directed by the reference. This reasonable expectation of success would motivate the artisan to modify Hardman to include freeze-drying steps after and/or before each heat treatment step.

16. Claims 36 and 42 are rejected under 35 U.S.C. 103(a) as being unpatentable over Hardman (US 3,620,919) in view of Yang (Shizhen Guoyi Guoyao (2006), vol. 17, no. 8, pp. 1503-4).

As discussed above, Hardman teaches a *T. foenum-graecum* extract obtained by incubating a mixture of the plant seeds in water over 3 hours; heating the mixture to reflux for up to 5 hours; and filtering the mixture to obtain a liquid extract. The reference teaches that the process is useful in extracting diosgenin from the *T. foenum-graecum*. However, the reference does not teach fractionating the extract using chromatography and a water/acetonitrile gradient. Yang teaches using HPLC with a water/acetonitrile mobile phase to recover diosgenin from *T. foenum-graecum* (see English abstract). Thus, an artisan of ordinary skill would reasonably expect that the HPLC method taught by Yang would be useful in recovering diosgenin from the diosgenin containing extract taught by Hardman. This reasonable expectation of success would motivate the artisan to modify Hardman to include the HPLC method taught by Yang. The

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artisan would be motivated to modify the parameters of the HPLC to maximize the recovery of the diosgenin.

17. No claims are allowed.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Susan Hoffman whose telephone number is (571)272-0963. The examiner can normally be reached on Monday-Thursday, 8:30-5:30.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Terry McKelvey can be reached on (571) 272-0775. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Susan Hoffman/
Primary Examiner, Art Unit 1655

Notice of References Cited	Application/Control No. 12/595,715	Applicant(s)/Patent Under Reexamination OLSEN, JENS STEEN	
	Examiner Susan Hoffman	Art Unit 1655	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-2005/0238738	10-2005	Lee et al.	424/757
	B	US-			
	C	US-			
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	Ziya Al-Din Abdullah Al-Jaam'e-li-Mufradaat-alAdvia-wal-Aghzia, Vol. II (13th century AD), Matba Amra, Cairo, Egypt, 1874 AD, page 25.
	V	Mohammad Akmal Khan Qaraabaadeen Azam wa Akmal (20th century AD), Matba Siddiqi, Delhi / Matba Mustafai, Delhi, 1909 AD, page 167
	W	Yang (Shizhen Guoyi Guoyao (2006), vol. 17, no. 8, pp. 1503-4).
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 12595715	Applicant(s)/Patent Under Reexamination OLSEN, JENS STEEN
	Examiner SUSAN HOFFMAN	Art Unit 1655

SEARCHED			
Class	Subclass	Date	Examiner
none	none		

SEARCH NOTES		
Search Notes	Date	Examiner
Searched all inventor names on PALM; USPT; PGPB; JPAB; EPAB; DWPI; EMBASE; MEDLINE; BIOSIS; CAPLUS; REG (see enclosed)	4/3/12	/sh/
STIC traditional knowledge and Medicine Historical and Classical Works (see enclosed list)	4/3/12	/s/h
Google search and TKDL search; keywords: fenugreek or Trigonella with freeze or freezing or froze or frozen	4/3/12	/sh/

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner

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